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F.2.5 Sample Exam Answers and Videos Questions 7-8

Question 7

0 points possible (ungraded)

7. Consider $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.

(a) (4 points) Show that $x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is one of its eigenvectors.

(b) (4 points) Find *ONE* eigenvalue of this matrix.

7. Consider $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.

(a) (4 points) Show that $x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is one of its eigenvectors.

Just multiply it out.

(b) (4 points) Find *ONE* eigenvalue of this matrix.

Either 0 (since the matrix clearly has linearly dependent columns) or 1 (the eigenvalue that corresponds to the given eigenvector.)

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Question 7

7. Consider $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

(a) (4 points) Show that $x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is one of its eigenvectors

$Ax = \lambda x$

$Ax = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

x is an eigenvector of A

(b) (4 points) Find *ONE* eigenvalue of this matrix

$\lambda = 1$
or $\lambda = 0$

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Question 8

0 points possible (ungraded)

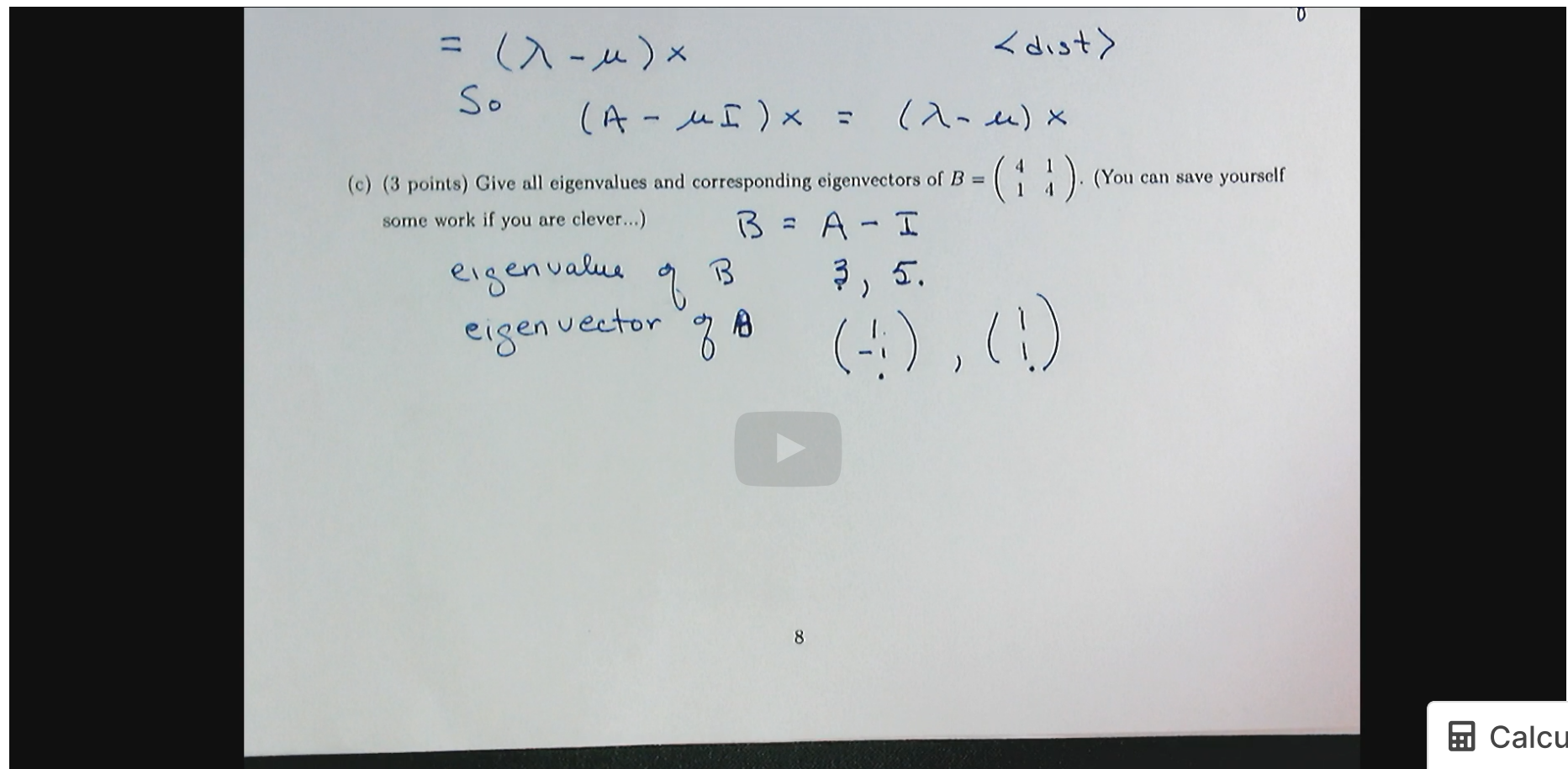
8. (a) (4 points) Give all eigenvalues and corresponding eigenvectors of $A = \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$.
- (b) (3 points) Let λ be an eigenvalue of A and $Ax = \lambda x$ so that x is an eigenvector corresponding to A . Show that $(A - \mu I)x = (\lambda - \mu)x$.
- (c) (3 points) Give all eigenvalues and corresponding eigenvectors of $B = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$. (You can save yourself some work if you are clever...)

8. (a) (4 points) Give all eigenvalues and corresponding eigenvectors of $A = \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$.
- $\det\left(\begin{pmatrix} 5-\lambda & 1 \\ 1 & 5-\lambda \end{pmatrix}\right) = (5-\lambda)^2 - 1 = 25 - 10\lambda + \lambda^2 - 1 = \lambda^2 - 10\lambda + 24 = (\lambda-4)(\lambda-6)$.
So, the eigenvalues are 4 and 6.
- Now, if $\lambda = 4$ then $\begin{pmatrix} 5-\lambda & 1 \\ 1 & 5-\lambda \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Clearly, the vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is in its null space and hence an eigenvector corresponding to $\lambda = 4$.
- Now, if $\lambda = 6$ then $\begin{pmatrix} 5-\lambda & 1 \\ 1 & 5-\lambda \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$. Clearly, the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is in its null space and hence an eigenvector corresponding to $\lambda = 6$.
- (b) (3 points) Let λ be an eigenvalue of A and $Ax = \lambda x$ so that x is an eigenvector corresponding to A . Show that $(A - \mu I)x = (\lambda - \mu)x$.
 $(A - \mu I)x = Ax - \mu Ix = \lambda x - \mu x = (\lambda - \mu)x$.
- (c) (3 points) Give all eigenvalues and corresponding eigenvectors of $B = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$. (You can save yourself some work if you are clever...)
- We know the eigenvalues and eigenvectors of $A = \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$ from part (a). But $B = A - 1 \times I$. So, by part (b) we know that the eigenvalues are 3 and 5 and the eigenvectors are $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

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Question 8



Handwritten solution for Question 8:

$= (\lambda - \mu)x$ <dist>

So $(A - \mu I)x = (\lambda - \mu)x$

(c) (3 points) Give all eigenvalues and corresponding eigenvectors of $B = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$. (You can save yourself some work if you are clever...)

$B = A - I$

eigenvalue of B 3, 5.

eigenvector of B $\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

8

Calculator

X 1.6

▶ 6:56 / 6:56

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